

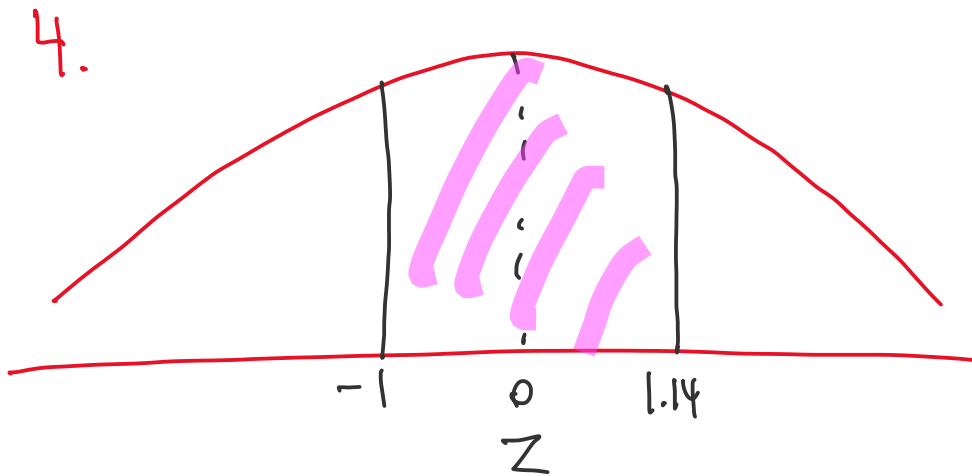
Thursday, July 9, 2026 1:56 PM

The Standard Normal Distribution

- To compute $P(0 \leq Z \leq z)$ can use
 - Minitab
 - A table
- Examples:
 - $P(0 \leq Z \leq 1.35) = ?$.4115
 - $P(Z > 1.46) = ?$ $1 - P(Z < 1.46) = 1 - .9279 = .0721$
 - $P(Z \leq -0.82) = ?$.2061
 - $P(-1 \leq Z \leq 1.14) = ?$ $.8729 - .1587 = .7142$
 - Find k such that $P(0 \leq Z \leq k) = 0.4$
 - Find m such that $P(-m \leq Z \leq m) = 0.6$

Statistics and Data Analysis – Summer 2026

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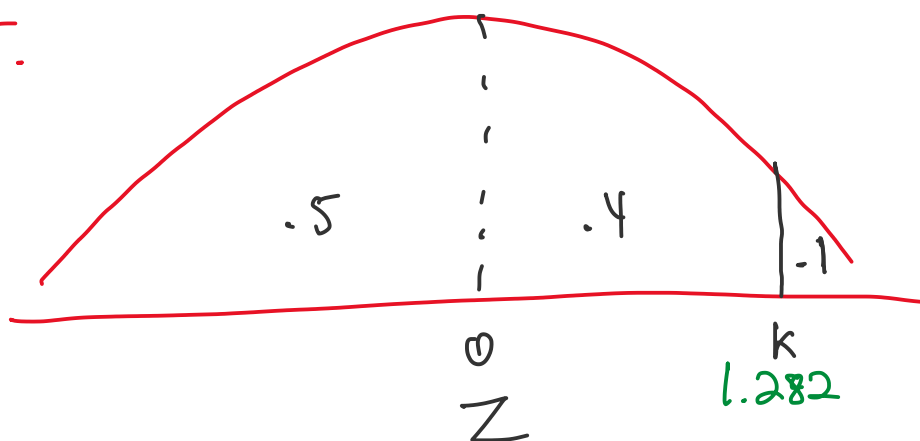


2.





5.



$$P(Z < k) = .9$$

Z	P
1.28	.8997
k	.9000
1.29	.9015

$$k \approx 1.28 + \frac{(.9000 - .8997)}{(.9015 - .8997)}$$

$$k \approx 1.28 + \left(\frac{3}{18}\right)(.01) \approx 1.28$$

Additional Interpolation Problem

$$P(Z < r) = .8 \text{ Find } r.$$

Z	P
.84	.7995
r	.8000
.85	.8023

$$r \approx .84 + \left(\frac{5}{28}\right)$$

Computing Probabilities – Example

- The daily demand for coke cans in a cafeteria is approximately normally distributed with mean 150 and standard deviation 20. Let X be a RV that represents the demand

1. $P(X \leq 200) = ?$

$$P\left(\frac{X-\mu}{\sigma} < \frac{200-150}{20}\right) = P(Z < 2.5) = .9938$$

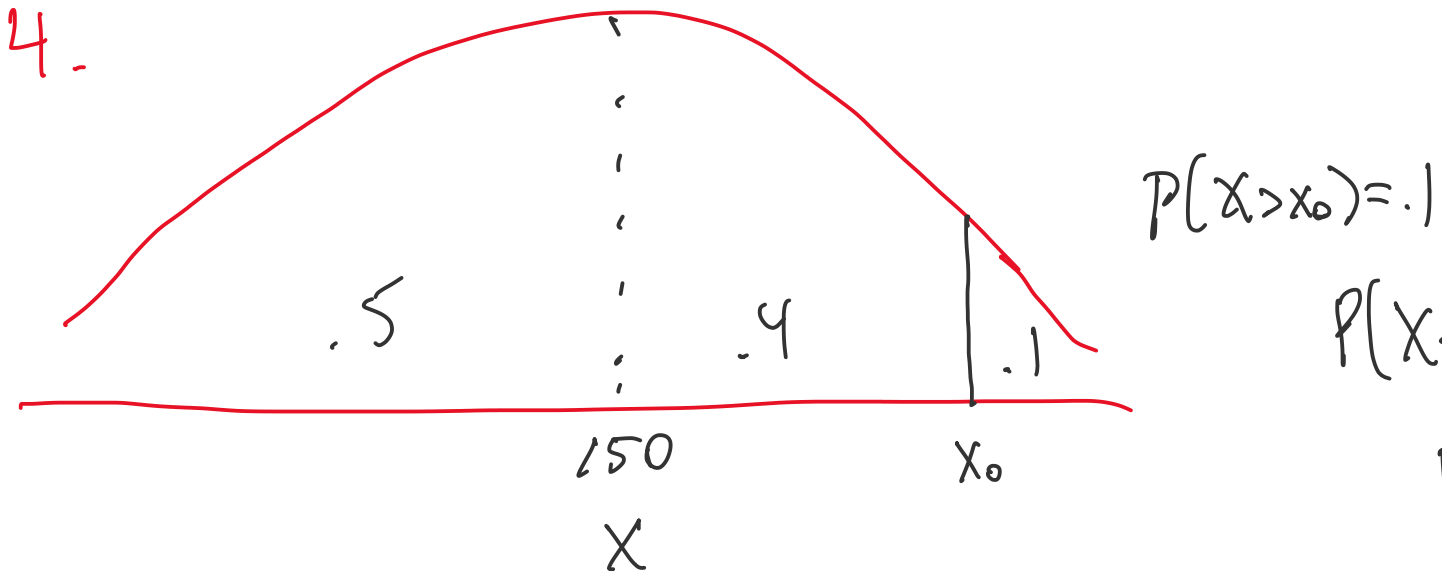
2. $P(X \geq 180) = ?$

$$1 - P(X < 180) = 1 - P\left(\frac{X-\mu}{\sigma} < \frac{180-150}{20}\right) = 1 - .9332 = .0668$$

3. $P(150 \leq X \leq 180) = ?$

$$.9332 - .5 = .4332$$

4. How many cans should be stocked so that $P(\text{running out of cans}) = 0.1$?



Random sample of n observations
iid: independent and identically distributed

$$E(X_i) = \mu$$

$$\text{Var}(X_i) = \sigma^2$$

$$\text{Remember } \bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

$$E(\bar{X}) = E\left[\sum_{i=1}^n \frac{X_i}{n}\right] = \sum_{i=1}^n E\left(\frac{X_i}{n}\right) = \sum_{i=1}^n \left(\frac{1}{n}\right) E(X_i)$$

$$\text{Var}(\bar{X}) = \text{Var}\left[\sum_{i=1}^n \frac{X_i}{n}\right] \stackrel{\text{IND}}{=} \sum_{i=1}^n \text{Var}\left(\frac{X_i}{n}\right) = \sum_{i=1}^n \left(\frac{1}{n}\right)^2 \text{Var}(X_i)$$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

Central Limit Theorem

Let $x_1, x_2, \dots, x_n$ be an iid random sample
 where $E(x_i) = \mu$ & $\text{Var}(x_i) = \sigma^2$
 and n is large ($n \geq 30$).

Then $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$, $\sum_{i=1}^n x_i \sim N(n\mu, n\sigma^2)$

Remember $\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$

Let's look at $\sum_{i=1}^n x_i$.

$$E[\sum x_i] = n\mu$$

$$\text{Var}[\sum x_i] = n\sigma^2$$

Consider $n=100$ one-ounce bags of chips.

$$E(x_i) = .95$$

$$\text{Var}(x_i) = (.02)^2$$

$$1. P(\bar{X} < .951) = P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \dots\right)$$

$$P(\bar{X} - \mu > \dots) = P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \dots\right)$$

var

$$\sigma_x = .02$$

$$2. P(X > .948) = 1 - P(X \leq .948)$$

$$3. P(-.948 < \bar{X} < .953) = P\left(\frac{-4}{.02} < Z < \frac{4}{.02}\right)$$

$$= P(-200 < Z < 200)$$

$$= P(-\infty < Z < \infty)$$

$$= .99$$

The Central Limit Theorem – Example

- The daily number of cars sold in a certain dealership has the following distribution:

x	0	1	2	3
P(X = x)	0.1	0.3	0.4	0.2

- Find $E(X)$ and $\text{Var}(X)$
- Assume 300 **days** in a year. $P(\text{Fewer than 495 cars sold in a year}) = ?$

